

AI 504 Problem set 6

Siddharth Bhaskar

Due Friday, May 15

Instructions. These are identical to preceding problem sets. Summary: you are allowed/encouraged to submit in groups; all students responsible for all problems; no adaptive online help. Give sufficient justification for all your answers, but avoid redundancy and/or scaffolding that helped you get to the answer, but is not necessary to include in the solution.

1. We are going to describe a translation from the syntax of $\mathcal{A}(\mathcal{RC})$ to the syntax of propositional logic. First of all, if (P, R) is a signature of $\mathcal{A}(\mathcal{RC})$ then just map it to the propositional logic signature $P \cup R$. (In other words, either a noun or a verb in my source signature can be used as a sentence letter in my target signature.)

Now map terms t of $\mathcal{A}(\mathcal{RC})$ to formulas t^* of propositional logic as follows. Each noun p becomes the propositional formula p . Then “ r all $(\dots)$ ” becomes “ $(\dots) \rightarrow r$,” so that, e.g., the term see all (love all dogs) becomes $(\text{dogs} \rightarrow \text{love}) \rightarrow \text{see}$. In the latter formula see, love, dogs are just boolean-valued sentence letters; we have forgotten the noun/verb distinction.

Finally, the translation of the sentence “All t s ” is $t^* \rightarrow s^*$, where t^* and s^* is the mapping on terms defined above. (Call this map $\varphi \mapsto \varphi^*$, reusing the symbol $\star$.) So, for example

$$\left(\text{All} (\text{see all} (\text{love all dogs})) (\text{love all birds}) \right)^* = ((\text{dogs} \rightarrow \text{love}) \rightarrow \text{see}) \rightarrow (\text{birds} \rightarrow \text{love}).$$

- (a) Suppose Γ is a set of sentences and φ is a sentence in $\mathcal{A}(\mathcal{RC})$, and suppose that $\Gamma \models \varphi$. Prove that $\Gamma^* \models \varphi^*$, as formulas of propositional logic.
 - (b) Show that the converse is not true. In other words, come up with a concrete sentences Γ, φ of $\mathcal{A}(\mathcal{RC})$ such that $\Gamma^* \models \varphi^*$ in propositional logic but $\Gamma \not\models \varphi$ in $\mathcal{A}(\mathcal{RC})$.
2. If φ is a sentence in propositional logic, let φ^+ be obtained from φ by negating all of the sentence letters. For example,

$$\left((p \vee (q \rightarrow r)) \leftrightarrow \neg(r \wedge p) \right)^+ = (\neg p \vee (\neg q \rightarrow \neg r)) \leftrightarrow \neg(\neg r \wedge \neg p).$$

- (a) For any satisfiable propositional formula φ , must φ^+ also be satisfiable? Either prove or give a counterexample.
 - (b) Can you find a formula φ such that φ is neither a tautology nor a contradiction and φ and φ^+ are logically equivalent? Find one or prove that none exists.
3. Work in the signature that has sentence letters $p_{i,j}$ for $0 \leq i, j \leq 9$ (one hundred letters in all). Consider the formula Φ which is the conjunction of all of the following clauses:

- $p_{0,9} \wedge p_{9,0}$,
- $p_{i,j} \rightarrow p_{i-1,j}$ for each $1 \leq i \leq 9$ and $0 \leq j \leq 9$, and
- $p_{i,j} \rightarrow p_{i,j-1}$ for each $0 \leq i \leq 9$ and $1 \leq j \leq 9$.

How many models satisfy Φ ? (*Hint.* If this seems too daunting, replace 9 with something smaller.)